

RouteReserve: Travel Booking & Management Platform

Software Requirements Specification (SRS) Document

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1. Introduction

1.1 Purpose: This document specifies the software requirements for the RouteReserve Unified Travel Booking & Management System. It describes the scope, system architecture, functional capabilities, user interfaces, database specifications, and non-functional requirements of the system.

1.2 Scope: RouteReserve is a next-generation unified travel agency portal designed to simplify routing reservations for flights, trains, buses, and car rentals. By connecting a Next.js frontend with a NestJS REST API backed by Prisma ORM and PostgreSQL, the platform allows customers to search transit options, customize passenger numbers, input traveler information, verify confirmation vouchers, and check their booking histories. The application specifically serves the Indian market by using Indian Rupees (INR) as the base currency with real-time USD conversion support.

2. Overall Description

2.1 Product Perspective: RouteReserve is structured as a full-stack monorepo consisting of: React 19 and Next.js 15+ (App Router) glassmorphic client interface, NestJS 11 REST API framework, and a PostgreSQL database connected via Prisma Client.

2.2 Product Functions

- **Unified Search Dashboard:** Select and scan transit schedules across flights, trains, buses, and cars.
- **Dynamic Flights Engine:** Toggle round-trip routes, adjust slider budgets, and calculate proximity surge fares based on departure dates.
- **Multiple Traveler Booking:** Book 1 to 9 passengers in a single transaction with automatic invoice pricing multiplication.
- **Interactive Admin Panel:** Visualize agency metrics, metrics charts, and booking ratios using Chart.js.

3. Specific Requirements

3.1 Functional Requirements

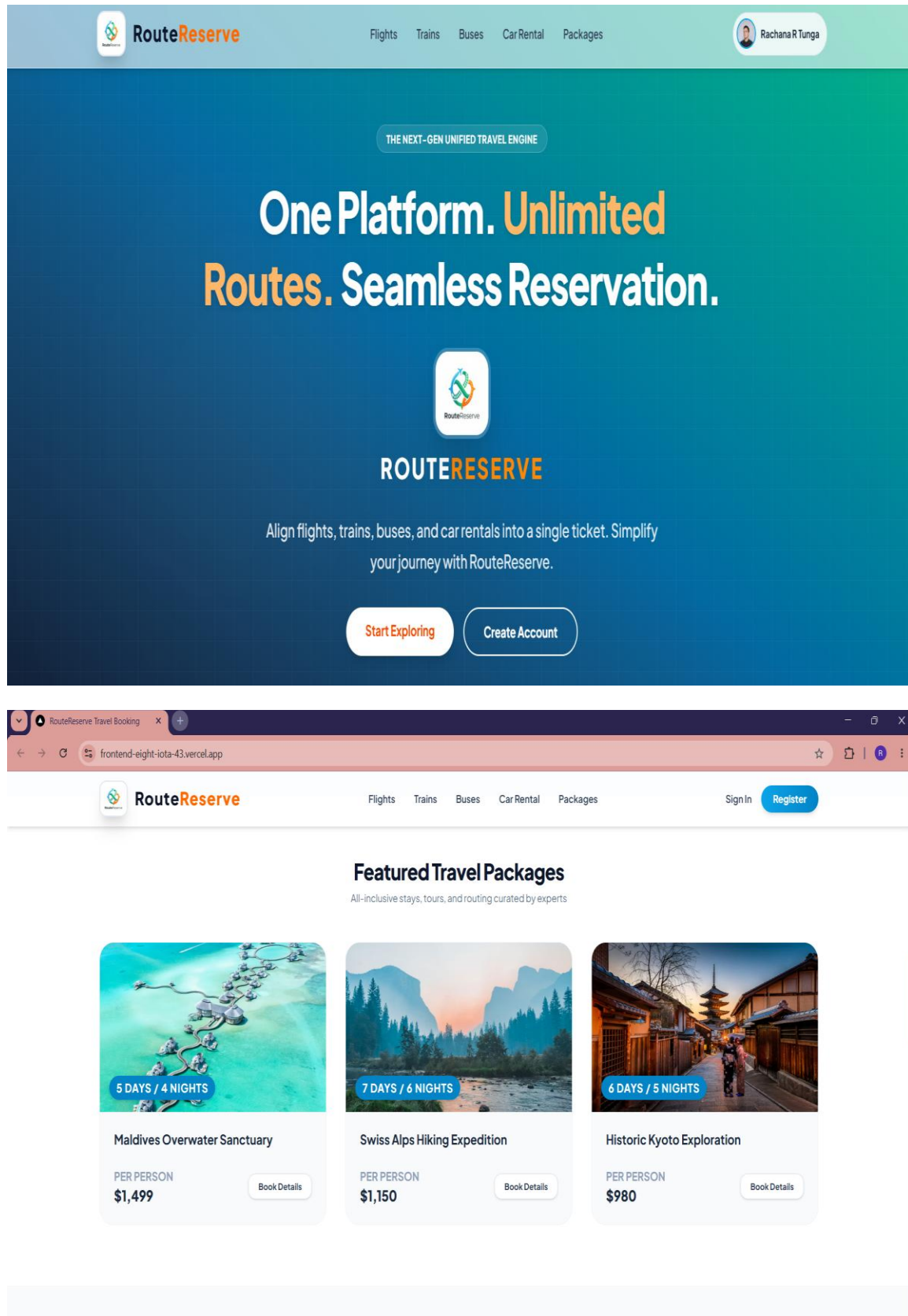
- **Flights Management Module:** Support One-Way and Round-Trip options. Selecting Round-Trip must unlock return date selectors and double the base ticket cost. Fares surge calculations automatically increase flight tickets if departure date is close.
- **Train Reservations Module:** Query 10+ train listings representing premium Indian Railway connections (Vande Bharat Express, Shatabdi, Rajdhani, local Mail/Express).
- **Bus Ticketing Module:** Query 20+ bus connection options representing private luxury sleepers (Suguma, Zingbus, etc.) and automatically double or triple base fares for longer inter-state transits.

4. System Schema & Database Specifications

The database layer maps entities such as User, Service, and Booking. The relationships are maintained using Prisma's referential constraints where every Booking belongs to a Service and holds details like customer Name, Email, Phone, booking date, and seat choices.

5. UI Screenshots Showcase

5.1 Landing Dashboard:





RouteReserve

Flights Trains Buses Car Rental Packages

Sign In

Register

Why Choose RouteReserve

Revolutionizing how multi-transit booking works



Unified Reservation Guarantee

Book your flight, train, and car in one click. If a flight delay causes you to miss a connection train, we reschedule it free of charge.



Dynamic Rates & Savings

By packaging air routes with train links and regional bus routes, our algorithm identifies savings up to 35% compared to purchasing individually.



24/7 Route Surveillance

Our operations team tracks all links in real-time. Receive automated notifications and dynamic routes changes on the fly.

150+

DESTINATIONS WORLDWIDE

1.2M+

HAPPY TRAVELERS

99.8%

ON-TIME ROUTING

45+

TRANSPORT PARTNERS



RouteReserve

Flights Trains Buses Car Rental Packages

Sign In

Register

What Our Travelers Say

Hear directly from verified RouteReserve members

★★★★★

"RouteReserve completely changed how I travel for work. Booking a flight, matching it with an express train, and hiring a car took under five minutes. The unified itinerary is a lifesaver."

S

Sarah Jenkins

BUSINESS CONSULTANT

★★★★★

"I love traveling off the beaten path. RouteReserve is the only booking site that handles small local bus systems and cross-country train connections seamlessly alongside international flights."

M

Marcus Vance

ADVENTURE PHOTOGRAPHER

Frequently Asked Questions

Need help? Browse our answers below.

5.2 Deployed Live Frontend Portal:

```
next-server (v16.2.10) X + v
Windows PowerShell 5.1
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\DELL> cd D:\RouteReserve-Travel-Booking-System
PS D:\RouteReserve-Travel-Booking-System> npm run dev:frontend

> routerreserve-monorepo@1.0.0 dev:frontend
> npm run dev --prefix frontend

> routerreserve-travel-booking-system@0.1.0 dev
> next dev --webpack

▲ Next.js 16.2.10 (webpack)
- Local:      http://localhost:3000
- Network:    http://192.168.56.1:3000
- Environments: .env.local
✓ Ready in 3.6s

o Compiling / ...
GET / 200 in 11.0s (next.js: 10.1s, application-code: 888ms)
o Compiling /dashboard ...
GET /dashboard 200 in 5.8s (next.js: 5.7s, application-code: 128ms)
GET /dashboard 200 in 51ms (next.js: 9ms, application-code: 42ms)
GET /dashboard 200 in 43ms (next.js: 6ms, application-code: 37ms)
GET / 200 in 592ms (next.js: 29ms, application-code: 563ms)
GET /bookings/flights 200 in 3.0s (next.js: 2.9s, application-code: 144ms)
GET /bookings/flights?from=Bangalore&to=Vietnam&date=2026-09-03&tripType=roundtrip&returnDate=2026-09-12 200 in 96ms (next.js: 17ms, application-code: 80ms)
)
GET /bookings/flights?from=Bangalore&to=Vietnam&date=2026-09-09&tripType=roundtrip&returnDate=2026-09-19 200 in 212ms (next.js: 96ms, application-code: 116ms)
GET /bookings/confirmation?id=BK-67993 200 in 2.4s (next.js: 2.4s, application-code: 67ms)
o Compiling /bookings/history ...
GET /bookings/history 200 in 3.6s (next.js: 3.4s, application-code: 176ms)
GET /bookings/history 200 in 72ms (next.js: 5ms, application-code: 67ms)
GET /bookings/trains 200 in 2.0s (next.js: 1980ms, application-code: 59ms)
GET /bookings/trains 200 in 51ms (next.js: 8ms, application-code: 43ms)
GET /bookings/flights 200 in 141ms (next.js: 49ms, application-code: 92ms)
GET /bookings/trains 200 in 39ms (next.js: 5ms, application-code: 34ms)
GET /bookings/trains?from=Barkur&to=KSR+Bengaluru&date=2026-09-13 200 in 47ms (next.js: 6ms, application-code: 41ms)
GET /bookings/confirmation?id=BK-95261 200 in 238ms (next.js: 79ms, application-code: 159ms)
```

5.3 Active Flights Scan (Vietnam & Hyderabad routes):

RouteReserve

FlightsTrainsBusesCar RentalPackages

Rachana R Tunga

Search Air Flights

Book your optimal global flight routes dynamically

One-Way Trip

Round-Trip

FROM

TO

DEPARTURE DATE

Q Re-Search

Hyderabad

California

28-07-2026

Filters

Reset

Stops

All Options

Direct Flight

1 Stop

Cabin Class

All Cabin Classes

Sort Options

Price: Lowest to Highest

Max Price (₹)

200000

INDIGO • 6E-100

Hyderabad (HYD) → California (CAL)

05:15 AM – 07:45 AM (1h 45m) 1 Stop

TICKET COST

₹16,200

Book Now

AIR INDIA • AI-187

Hyderabad (HYD) → California (CAL)

07:45 AM – 09:15 AM (2h 50m) Direct

TICKET COST

₹17,496

Book Now

VISTARA • UK-274

Hyderabad (HYD) → California (CAL)

09:15 AM – 11:45 AM (1h 55m) Direct

TICKET COST

₹18,792

Book Now

RouteReserve

FlightsTrainsBusesCar RentalPackages

Rachana R Tunga

Search Air Flights

Book your optimal global flight routes dynamically

One-Way Trip

Round-Trip

FROM

TO

START DATE

END DATE (RETURN)

Q Re-Search

Bangalore

Vietnam

09-09-2026

19-09-2026

Filters

Reset

Stops

All Options

Direct Flight

1 Stop

Cabin Class

All Cabin Classes

Sort Options

Price: Lowest to Highest

Max Price (₹)

200000

SKYLINE AIRWAYS • SA-100

Bangalore (BAN) → Vietnam (VIE)

05:15 AM – 04:45 PM (10h 15m) 1 Stop

ROUND-TRIP PRICE

₹1,08,000 (\$1,301)

Incl. Return Flight

Book Now

AIR INDIA • AI-187

Bangalore (BAN) → Vietnam (VIE)

07:45 AM – 06:15 PM (11h 30m) 1 Stop

ROUND-TRIP PRICE

₹1,16,640 (\$1,405)

Incl. Return Flight

Book Now

SWISS INTERNATIONAL • LX-274

Bangalore (BAN) → Vietnam (VIE)

09:15 AM – 08:45 PM (12h 45m) Direct

ROUND-TRIP PRICE

₹1,25,280 (\$1,509)

Incl. Return Flight

Book Now

EMIRATES • EK-101

ROUND-TRIP PRICE

5.4 Passenger Details Registration Modal:

FlightsTrainsBusesCar RentalPackages

✈️

Complete Route Reservation

✕

SWISS INTERNATIONAL

CLASS: ECONOMY

OUTBOUND

Bangalore (BAN) → Vietnam (VIE)

Date: 2026-09-09

Departure: 09:15 AM

RETURN

Vietnam (VIE) → Bangalore (BAN)

Date: 2026-09-19

Departure: 04:30 PM (est)

NUMBER OF PASSENGERS

1 Passenger

PASSENGER 1 NAME

Full Name as on ID

AGE

Age

SELECT CLASS

Economy

CHOOSE SEAT

12A

TOTAL COST

₹1,25,280 (\$1,509)

Secure Purchase

RouteReserve

Search Train Tickets

Book express or scenic train connections seamlessly

DEPARTURE STATION

Barkur

Filters

Reset

Coach Class

All Cabin Classes

Sort Options

Price: Lowest to Highest

Max Ticket Price (₹)

10000

FlightsTrainsBusesCar RentalPackages

✈️

Book Rail Ticket

✕

MAIL EXPRESS

CLASS: BUSINESS

Barkur → KSR Bengaluru

Date: 2026-09-13

Time: 06:00 AM

NUMBER OF PASSENGERS

2 Passengers

PASSENGER 1 NAME

Rachana R Tunga

AGE

24

PASSENGER 2 NAME

N Ramachandra

AGE

67

SELECT CLASS

Business

SELECT CARRIAGE & SEAT

Car 1, Seat 14C

TOTAL COST

₹980

Secure Purchase

DEPARTURE STATION

KSR

Search Trains

TICKET COST

₹490

Book Now

TICKET COST

₹608

Book Now

TICKET COST

₹798

Book Now

08:30 AM - 01:15 PM (6h 15m)

AC 3 Tier

 **Book Bus Ticket**

BHARATI TRAVELS

Bangalore → Shimogga

Date: 2026-07-18

AC SLEEPER

Time: 06:00 AM

NUMBER OF PASSENGERS

2 Passengers

PASSENGER 1 NAME

Divya Ram

AGE

45

PASSENGER 2 NAME

Akanksha Upadhya

AGE

15

SELECT BUS SEAT

Seat 18 (Window)

TOTAL COST

₹1,900

 **Secure Purchase**

 **RouteReserve**

Flights Trains Buses Car Rental Packages

 Rachana R Tunga

 **Book Vehicle Rental**

AVIS INDIA

TOYOTA

Automatic

Toyota Toyota Innova Crysta (Luxury MUV)

Rate: ₹6,000/day

Location: Heathrow Airport Terminals (LHR)

LEAD DRIVER FULL NAME

Rachana R Tunga

PICKUP DEPOT ADDRESS

Heathrow Airport Terminals (LHR)

RENTAL DURATION (DAYS)

3

ESTIMATED COST (3 DAYS)

₹18,000

 **Secure Purchase**



4.9

LUXURY MUV

Toyota Innova Crysta

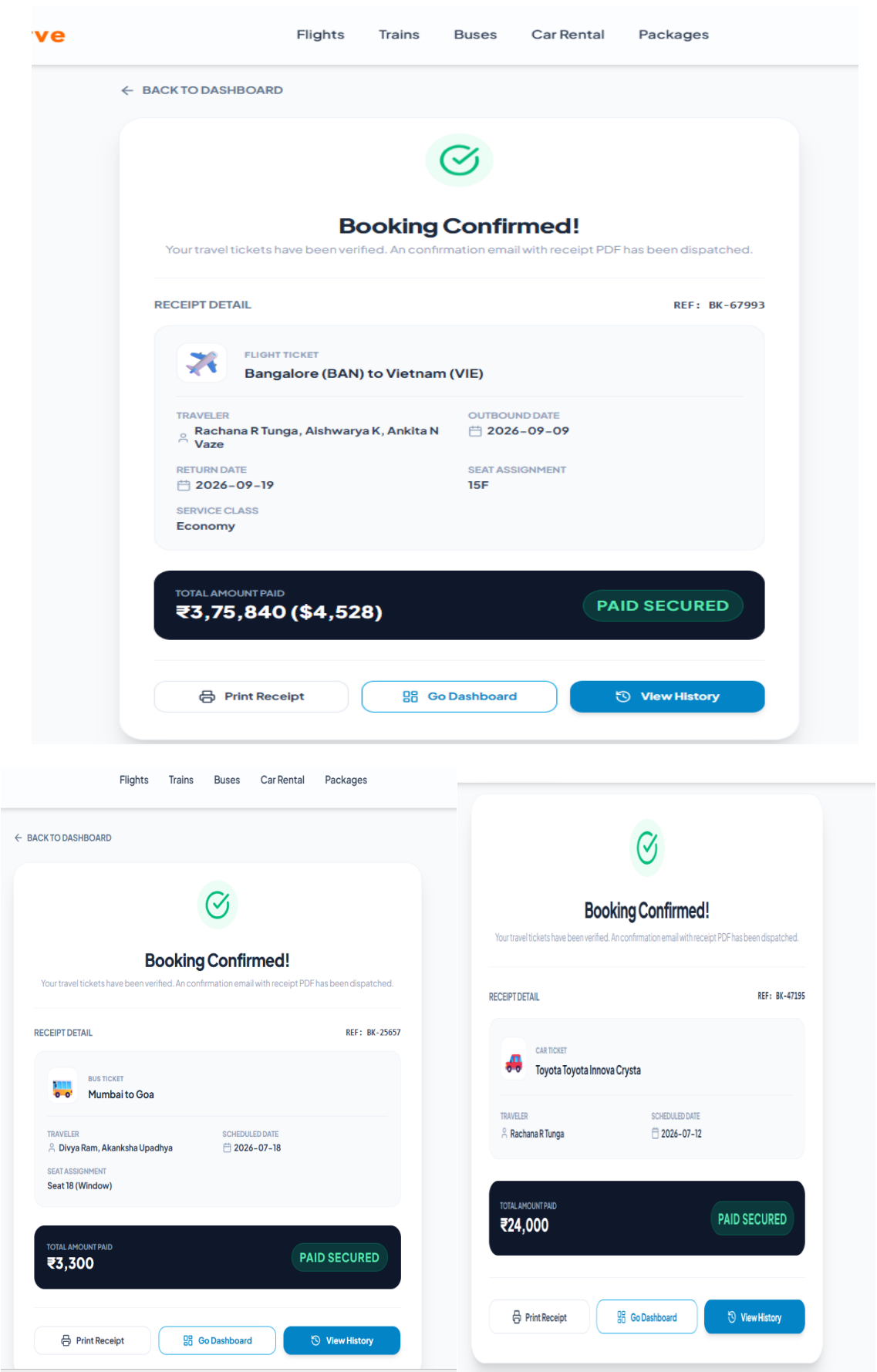
Automatic



 **RouteReserve**

COMPANY SERVICES SUPPORT

5.5 Confirmed Invoice Receipt Voucher:



5.6 PDF Generated Ticket Download File:

The screenshot shows a web browser window with the URL `localhost:3000/bookings/confirmation?id=BK-67993`. The main content area displays a 'Booking Confirmed!' message with a green checkmark icon. Below this, it states: 'Your travel tickets have been verified. An confirmation email with receipt PDF has been dispatched.'

The 'RECEIPT DETAIL' section includes the following information:

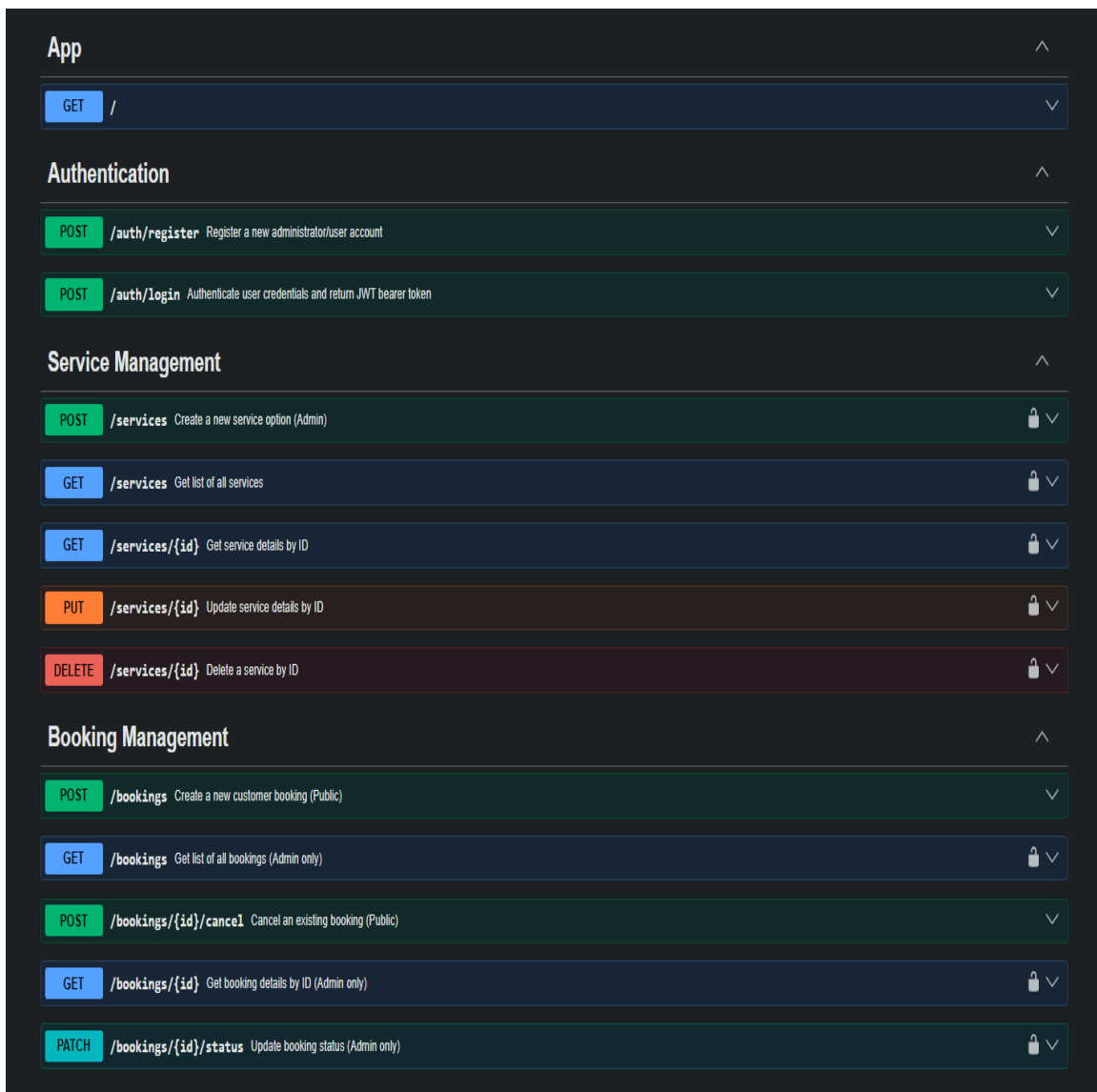
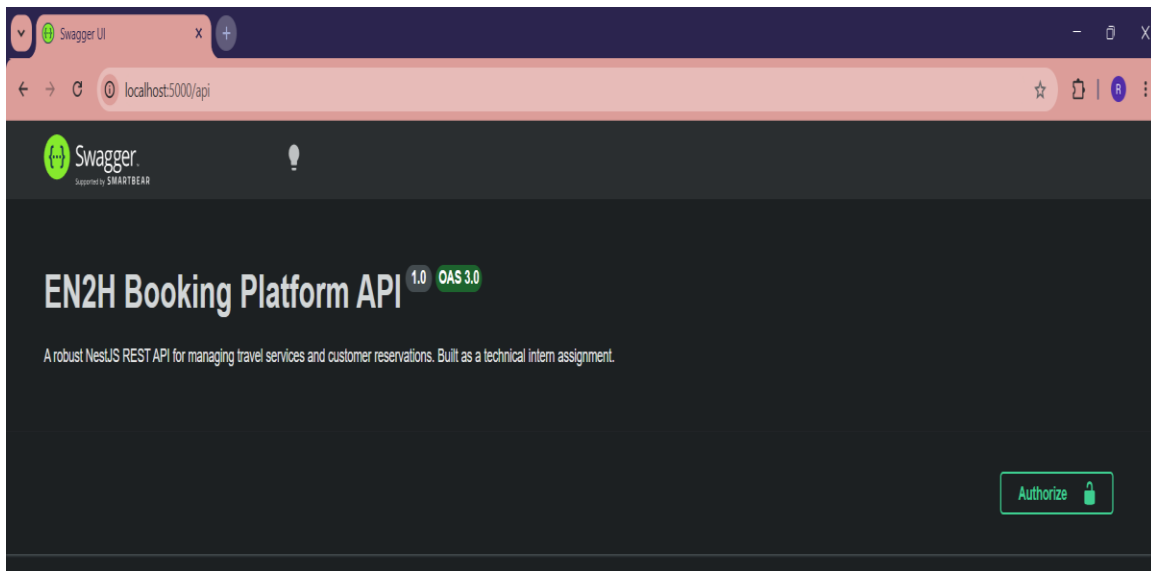
- REF: BK-67993
- FLIGHT TICKET: Bangalore (BAN) to Vietnam (VIE)
- TRAVELER: Rachana R Tunga, Ashwarya K, Anika N, Vase
- OUTBOUND DATE: 2024-09-09
- RETURN DATE: 2024-09-19
- SEAT ASSIGNMENT: 15F
- SERVICE CLASS: Economy

The 'TOTAL AMOUNT PAID' is ₹3,75,840 (\$4,528), and the status is 'PAID SECURED'.

At the bottom of the receipt, there are three buttons: 'Print Receipt', 'Go Dashboard', and 'View History'.

A 'Print' overlay is visible on the right side of the screen, showing options to 'Save as PDF', 'All' pages, and 'Portrait' layout. The overlay also includes a 'More settings' dropdown and 'Save' and 'Cancel' buttons at the bottom.

5.7 Swagger OpenAPI interactive Developer Panel:



Booking Management

POST /bookings Create a new customer booking (Public)

Parameters

Cancel

Reset

No parameters

Request body ^{required}

application/json

Edit Value | Schema

```
{
  "customerName": "Rachana R Tunga",
  "customerEmail": "user1@gmail.com",
  "customerPhone": "+919876543210",
  "serviceId": 30,
  "bookingDate": "2026-08-15",
  "bookingTime": "14:30",
  "notes": "Need window seat if possible"
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:5000/bookings' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "customerName": "Rachana R Tunga",
    "customerEmail": "user1@gmail.com",
    "customerPhone": "+919876543210",
    "serviceId": 30,
    "bookingDate": "2026-08-15",
    "bookingTime": "14:30",
    "notes": "Need window seat if possible"
  }'
```

Request URL

http://localhost:5000/bookings

Server response

Code Details

201

Response body

```
{
  "id": 9,
  "customerName": "Rachana R Tunga",
  "customerEmail": "user1@gmail.com",
  "customerPhone": "+919876543210",
  "serviceId": 30,
  "bookingDate": "2026-08-15T00:00:00.000Z",
  "bookingTime": "14:30",
  "status": "PENDING",
  "notes": "Need window seat if possible",
  "createdAt": "2026-07-13T17:06:07.716Z",
  "updatedAt": "2026-07-13T17:06:07.716Z",
  "service": {
    "id": 30,
    "title": "Bullet Train (Kyoto Route)",
    "description": "Premium Shinkansen rail booking with green carriage privileges.",
    "duration": 180,
    "price": 150,
    "isActive": true,
    "createdAt": "2026-07-13T16:43:36.954Z",
    "updatedAt": "2026-07-13T16:43:36.954Z"
  }
}
```

 Download

Response headers

```
access-control-allow-origin: *
connection: keep-alive
content-length: 579
content-type: application/json; charset=utf-8
date: Mon, 13 Jul 2026 17:06:07 GMT
etag: W/"243-5+4A197o8IXvp3Q1N7XlhxV9s6s"
keep-alive: timeout=5
x-powered-by: Express
```

Responses

Code	Description	Links
201	Booking successfully created.	No links
400	Invalid date or inactive service.	No links
409	Service slot duplicate conflict.	No links

